16/10/22

$c_{n+2} a_{n+2} + c_{n+1} a_{n+1} + c_n a_n = f(n)$

$f(n) = a_t n^t + a_{t-1} n^{t-1} + \dots + a_1 n + a_0$

Eg: $S_n = \sum_{i=1}^{n} i^2 \quad S_1 = 1$

$S_n = S_{n-1} + n^2$ if, open it up,

$f(n) = n^2$

$S_n^{(h)} = A(1)^n$

$S_n^{(p)} = B_2 n^2 + B_1 n + B_0$ → n^2 dominating

conflicts with A.

then raise the power of $S_n^{(p)}$.

$(B_2 n^2 + B_1 n + B_0) n$

$B_2 n^3 + B_1 n^2 + B_0 n = B_2(n-1)^3 + B_1(n-1)^2 + B_0(n-1) + n^2$.

Coeff. n^2 :
n :
const :

	$f(n)$	$a_n^{(p)}$
①	dr^n	Ar^n (no conflict with $a_n^{(h)}$)
②	$a_k n^k + a_1 n + a_0$	$A_t n^t + \dots + A_1 n + A_0$ (conflict raise powers)
③	$\sin n\theta$ / $\cos n\theta$	$A_1 \sin n\theta + A_2 \cos n\theta$
④	$r^n(\sin n\theta)$	$Ar^n(A_1 \sin n\theta + A_2 \cos n\theta)$.
⑤	$r^n(a_t n^t + \dots a_1 n + a_0)$	$Ar^n(A_t n^t + \dots + A_1 n + A_0)$.

General way of solving:-

$f(n) = \sum_{i=0}^{\infty} a_i x^i$ ← generating function

$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$.

${}^nC_0 \quad {}^nC_1 \quad {}^nC_2 \quad \dots \quad {}^nC_n \quad 0\ 0\ 0$

$\frac{1}{1-x} = 1 + x + x^2 + \dots$

$1 \quad 1 \quad 1 \quad \dots$

$$\frac{1}{1-x^2} = 1 + x^2 + x^4 + x^6 \cdots$$

$$1\ 0\ 1\ 0\ 1\ 0\ 1 \cdots$$

$a_i \rightarrow$ coefficients generated by function.
↳ sequence.

$a_n \rightarrow$ then coeff. of x^n

$$a_{n+2} - 5a_{n+1} + 6a_n = 2 \quad \forall n \geq 0,$$

$$a_0 = 3$$
$$a_1 = 7$$

Assume that, $a_0, a_1 \cdots a_n$

a's generated by $f(x) = \sum_{i=0}^{\infty} a_i x^i$

~~$6x^2 - 6x + 1$~~
~~$6x$~~

$10x^2 - 5x - 6x + 3$

$5x(2x-1) - 3(2x-1)$

$$f(x) = \frac{5x-3}{(1-x)(3x-1)} = \frac{A}{1-x} + \frac{B}{3x-1}$$

$$= \left(\frac{1}{1-x}\right) + \left(\frac{2}{1-3x}\right) \quad 3A - B = 5 \qquad A + B = -3$$

Coeff. of x^n in $2(1+(3x)^2 + \cdots$ $\quad 4A = 2$ $\quad B = -\frac{7}{2}$

$1 + x + x^2$

$$A = \frac{1}{2}$$

$$\boxed{a_n = 1 + 2 \cdot 3^n}$$

$^nC_0$ $^nC_1$ $^nC_2$ --- ^nC_n 0 0 ---

$\frac{1}{1-x} = 1+x+x^2+\cdots$ genarates Coeff 1 1 1 --

$\frac{1}{1-x^2} = 1+x^2+x^4+x^6+\cdots$ generates Coeff 1 0 1 0 1 --

Ex : $a_{n+2} - 5a_{n+1} + 6a_n = 2 \quad \forall n \geq 0 \quad a_0 = 3 \quad a_1 = 7$

Assume that $a_0, a_1, \cdots, a_n, \cdots$ is generated by $f(x) = \sum_{i=0}^{\infty} a_i x^i$

Sol :

$$\sum_{n=0}^{\infty} a_{n+2}x^{n+2} - 5a_{n+1}x^{n+2} + 6a_n x^{n+2} = \sum_{n=0}^{\infty} 2x^{n+2}$$

$$\sum_{n=0}^{\infty} a_{n+2}x^{n+2} - 5x\sum_{n=0}^{\infty} a_{n+1}x^{n+1} + 6x^2\sum_{n=0}^{\infty} a_n x^n = 2x^2\sum_{n=0}^{\infty} x^n$$

$$(f(x) - a_0 - a_1x) - 5x(f(x) - a_0) + 6x^2 f(x) = \frac{2x^2}{1-x}$$

$$(f(x))(6x^2 - 5x + 1) - a_0 + (5a_0 - a_1)x = \frac{2x^2}{1-x}$$

$$f(x)(2x-1)(3x-1) = \frac{2x^2}{1-x} + 3 - 8x = \frac{2x^2 + 3 - 3x - 8x + 8x^2}{1-x}$$

$$= \frac{10x^2 - 11x + 3}{1-x} = \frac{(5x-3)(2x-1)}{1-x}$$

$$f(x) = \frac{5x-3}{(1-x)(3x-1)} = \frac{1}{1-x} + \frac{2}{1-3x}$$

x^n Coeff from series expansions $a_n = 1 + 2.3^n$

a_n

$a(n,r)$ = # ways of selecting with repitition r items from n distinct items.

Combinatorial

~~$a(0,r) = 0 \quad \forall\ n \geq 0$~~

$a(n,0) = 1 \quad \forall\ n \geq 1$

$a(0,r) = 0 \quad \forall\ r > 0$.

$a(n,r) = a(n-1,r) + a(n,r-1)$ (r: 'with repetition') (look at last object)

not picked (?) picked (?)

$\forall\ n \in \mathbb{N}$ $f_n(x) = \sum_{r=0}^{\infty} a(n,r)\, x^r$

$r \geq 1$ $\sum_{r=1}^{\infty} a(n,r)\, x^r = \sum_{r=1}^{\infty} a(n-1,r)\, x^r + \sum_{r=1}^{\infty} a(n,r-1)\, x^r$

$\{f_n(x) - a(n,0)\} = \{f_{n-1}(x) - a(n-1,0)\}$

$+ x f_n(x)$.

$\Rightarrow f_n(x)(1-x) = f_{n-1}(x)$

$\Rightarrow f_n(x) = \frac{f_{n-1}(x)}{1-x} = \frac{f_{n-2}(x)}{(1-x)^2}$

$= \frac{f_0(x)}{(1-x)^n}$

$= \frac{1}{(1-x)^n}$

$= (1-x)^{-n}$

$a_n \rightarrow$ Coefficient of x^n in $(1-x)^{-n} \rightarrow \binom{n+r-1}{r}$

(→) $b(n,r)$ = # ways of selecting w/o repition r items from n distinct items.

$b(n,0) = 1 \quad \forall\ n \geq 0$ (look at last object)

$b(0,r) = 0 \quad \forall\ r > 0$

$r \geq 1$, $b(n,r) = b(n-1,r-1) + b(n-1,r)$

$\forall\ n \in \mathbb{N}$ $g_n(x) = \sum_{r=0}^{\infty} b_n(n,r)\, x^r$

$$b(n,r)\,x^r = b(n-1,r-1)\,x^r + b(n-1,r)\,x^r$$

$$\Rightarrow \sum_{r=0}^{\infty} b(n,r)\,x^r = x\cdot\sum_{r=0}^{\infty} b(n-1,r-1)\,x^{r-1} + \sum_{r=0}^{\infty} b(n-1,r)\,x^r$$

$$f_n(x) - 1 = x\,f_{n-1}(x) + f_{n-1}(x) - 1$$

$$f_n(x) = (1+x)\,f_{n-1}(x)$$

$$= (1+x)^2 f_{n-2}(x)$$

$$= (1+x)^n f_0(x)$$

$$f_n(x) = (1+x)^n$$

$$b(n,r) = {}^nC_r$$

(a) $e_1 + e_2 + e_3 + e_4 = n$

s.t e_1 is even $\quad (1+x^2+x^4+\cdots) = \frac{1}{1-x^2}$

e_2 is odd $\quad (x+x^3+x^5\cdots) = \frac{x}{1-x^2}$

$0 \le e_3 \le 4 \quad (1+x+x^2+\cdots+x^4) = \frac{1-x^5}{1-x}$

$e_4 \ge 1 \quad (x+x^2+\cdots) = \frac{x}{1-x}$

$$x^{e_1}\cdot x^{e_2}\cdot x^{e_3}\cdot x^{e_4} = x^n.$$

$$g(x) = \frac{(1-x)^5\,(x^2)}{(1-x^4)^2(1-x)^2} \rightarrow \text{generating function for } a_n$$

Combinations:

$C(3)$: 3
$1+1+1$
$1+2$
$2+1$ } non-zero summands

$C(n) \to 2^{n-1}$ closed form.

$p(n)$: $p(3)$ 3
$1+1+1$
$1+2$ partition

$P_0 = 1$ (for def^n)
$P_1 = 1$
$P_2 = 2$
$P_3 = 3$

$$g(x) = \sum_{n=0}^{\infty} P_n x^n$$

$1 \to (1+x+x^2+\cdots) \to \frac{1}{1-x}$
$2 \to (1+x^2+x^4-\cdots) \to \frac{1}{1-x^2}$

$g(x)$ coeff $x^n \to \left(\frac{1}{1-x}\right)\left(\frac{1}{1-x^2}\right)\cdots$

$$\prod_{r=1}^{\infty} \frac{1}{(1-x^r)}$$

$P_d(n)$ = # ways of selecting non-zero summands adding to n st each summand is distinct //

$P_d(1) = 1$ $\quad P_d(0) = 1$
$P_d(2) = 1$.
$P_d(3) = 2$

$$g_d(x) = \sum_{n=0}^{\infty} P_d^n x^n \to \left(\frac{1-x^2}{1-x}\right)\left(\frac{1-x^4}{1-x^2}\right)\left(\frac{1-x^6}{1-x^3}\right)\left(\frac{1-x^8}{1-x^4}\right)$$

$(1+x)(1+x^2)\cdots$

$P_o(n)$ = # ways of selecting non-zero summands adding to n such that each summand is odd

$$g_o(x) = \sum_{n=0}^{\infty} P_o^n x^n.$$

$P_o(0) = 1$
$P_o(1) = 1$
$P_o(2) = 1$
$P_o(3) = 2$

$(1+x+x^2+\cdots)(1+x^3+x^6+\cdots)(1+x^5+x^{10}+\cdots)$

$\frac{1}{1-x}$ $\quad \frac{1}{1-x^3}$ $\quad \frac{1}{1-x^5}$

$g_d(n) = g_b(n)$ //

$$(1+x)^n = \frac{n!}{0!\,n!} + \frac{n!}{1!\,(n-1)!}x + \frac{n!}{2!\,(n-2)!}x^2 + \cdots + \frac{n!}{n!\,0!}x^n$$

$$= {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$$

$$= {}^nP_0 \cdot x + {}^nP_1 \frac{x}{1!} + {}^nP_2 \frac{x^2}{2!} + \cdots + {}^nP_n \frac{x^n}{n!}$$

← exponential gen. function

$$g(n) = \sum_{n=0}^{\infty} a_n \frac{x^n}{n!}$$

well known

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!}$$

$$\frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$$

$$\frac{e^x - e^{-x}}{2} = \frac{x^1}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$$

Can we do in generating function approach?

Eg: 11 new employees into 4 divisions → (A, B, C, D)

s.t each division gets atleast 1 new employee.

Sol:

$$\left(\frac{x}{1!} + \frac{x^2}{2!} + \cdots\right)^4 = \left(e^x - 1\right)^4$$

Read coeff. of $\frac{x^{11}}{11!}$

$$= e^{4x} - 4e^{3x} + 6e^{2x} - 4e^x + 1$$

Coeff. of $\frac{x^{11}}{11!}$

$$= 11!\left(\frac{4^{11}}{11!} - 4\cdot\frac{3^{11}}{11!} + \frac{6\cdot 2^{11}}{11!} - 4\,\frac{1^{11}}{11!} + 0\right)$$

$$= 4^{11} - 4\cdot 3^{11} + 6\cdot 2^{11} - 4\cdot 1$$

$$f(x) = \sum_{n=0}^{\infty} a_n x^n.$$

$$\frac{f(x)}{1-x} = (a_0 + a_1 x + a_2 x^2 \cdots)(1 + x + x^2 + \cdots)$$

$$= b_0 + b_1 x + b_2 x^2 + \cdots + b_i x^i$$

$$(a_0) \quad (a_0 + a_1) \quad (a_0 + a_1 + a_2) \quad \sum_{i=0}^{i} a_i$$

$x + x^2 \rightarrow 0, 1, 1, 0, 0, \ldots$

$\frac{(x+x^3)}{(1-x)} \rightarrow 0, 1, 2, 2, 2, 2, \ldots$

$\frac{(x+x^2)}{(1-x)^2} \rightarrow 0, 1, 3, 5, 7, 9, 11, \ldots$

$\frac{(x+x^2)}{(1-x)^2} \rightarrow$

$$\sum_{k=1}^{i+1} (2k-1) = i^2$$

Coeff. of x^i in $x(1-x)^{-3} + x^2(1-x)^{-3}$.

Coeff. of x^{i-1} in $(1-x)^{-3}$ + $\binom{3+i-2}{i-1}$

Coeff. of x^{i-2} in $(1-x)^{-3}$ $\binom{3+i-3}{i-2}$

$$= \binom{i+1}{i-1} + \binom{i}{i-2}$$

$$\sum_{k=1}^{i+1} (2k-1) = i^2$$

can also be proved by gen. functions.